CYBERSECURITY INTERSHIP REPORT

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SUBMITTEB TO: ARCH TECHNOLOGIES

Month1 Assignment

TASK 1:

BASIC NETWORK SNIFFER

Build a network sniffer in Python that captures and analyzes network traffic. This project will help you understand how data flows on a network and how network packets are structured.

1. Overview

I developed a Python-based Network Sniffer to capture and analyze live network traffic. This project demonstrates how data packets move across a network by extracting Source and Destination IP addresses.

2. Tools Used

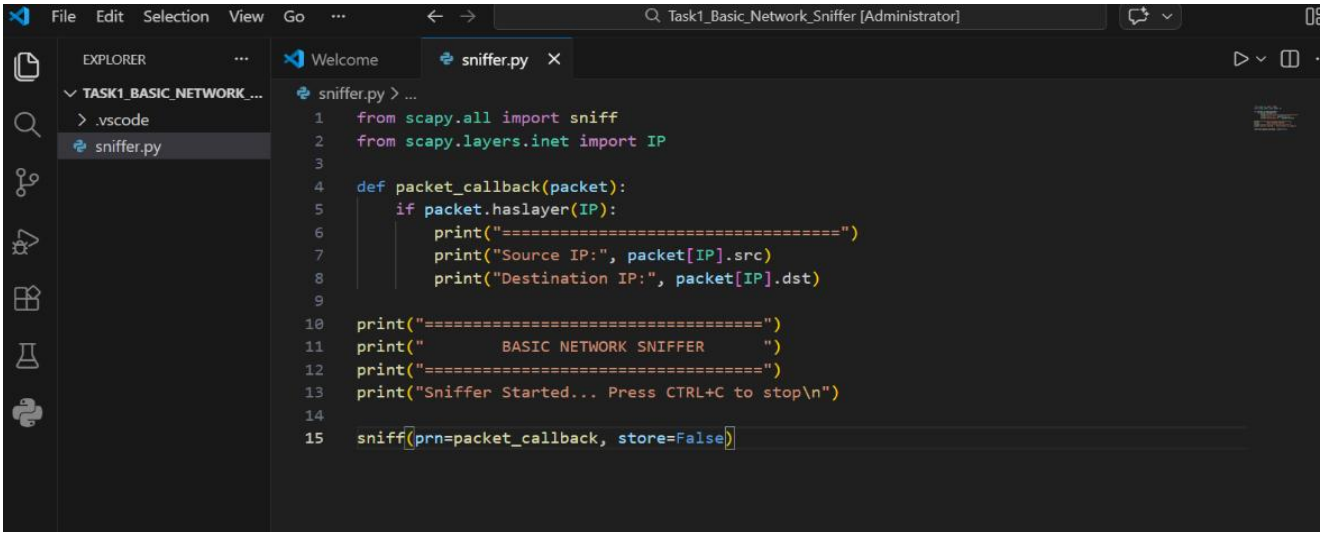
- **Language:** Python 3.11
- **Library: Scapy** (for packet sniffing and parsing)
- **Editor:** VS Code
- **Environment:** Windows PowerShell (Run as Administrator)

3. How I Completed the Task

1. **Environment Setup:** Installed the Scapy library and configured the Python environment in VS Code.
2. **Coding the Logic:**
    - Imported sniff and IP modules from Scapy.
    - Defined a callback function (packet_callback) to check every captured packet for an IP layer.

- Extracted and formatted the packet [IP].src and packet [IP]. dst fields for display.

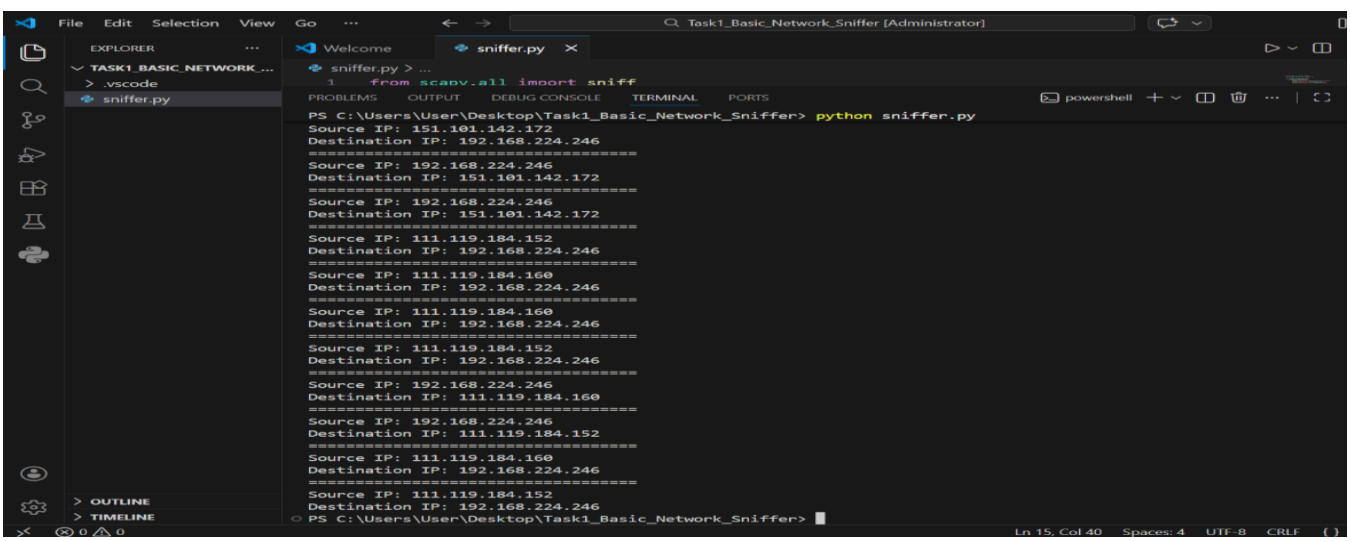
3. **Running the Sniffer:** Initiated the sniff () function with the callback and store=False to handle real-time traffic efficiently without consuming excessive memory.



4. Results & Analysis

As shown in my screenshots:

- The script successfully captures live traffic on my local network.
- It identifies communication between my machine (192.168.224.246) and external servers (e.g., 151.101.142.172) in real-time.



5. Conclusion

The task was successfully completed. I gained hands-on experience in packet interception and understanding the structure of the Network Layer (IP).

TASK 2:

KEY LOGGING

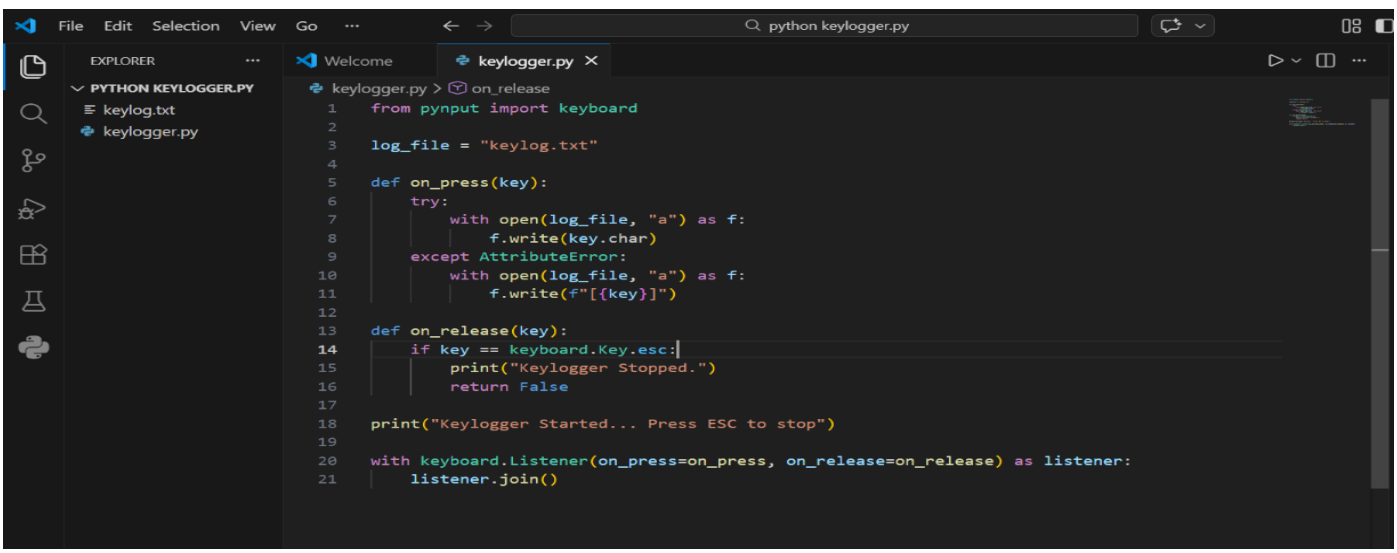
Simulate a basic keylogger in a controlled and safe environment to understand how keystroke logging works. Capture user keystrokes locally using a simple script, log them into a file, and analyze the potential risks associated with such attacks.

1. Overview

The objective of this task was to simulate a basic Keylogger in a controlled environment. This project demonstrates how user keystrokes can be intercepted and recorded, highlighting the critical risks to personal data and the importance of system-level security.

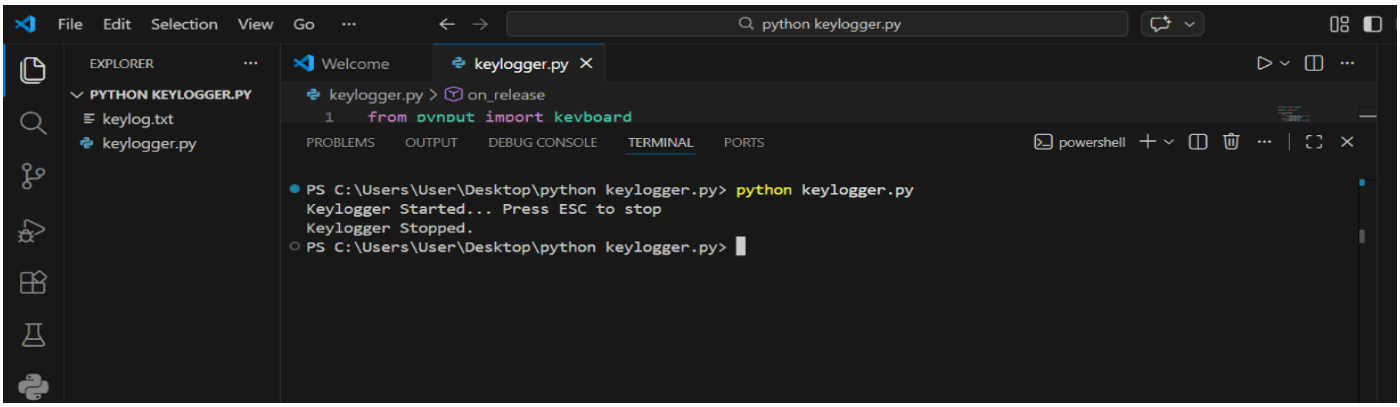
2. Tools & Technologies

- **Language:** Python 3.11
- **Library:** pynput (Specifically the keyboard module for monitoring events).
- **Editor:** Visual Studio Code (VS Code).
- **Log File:** keylog.txt (Stored within the "Desktop Files" directory).



3. How I Completed the Task

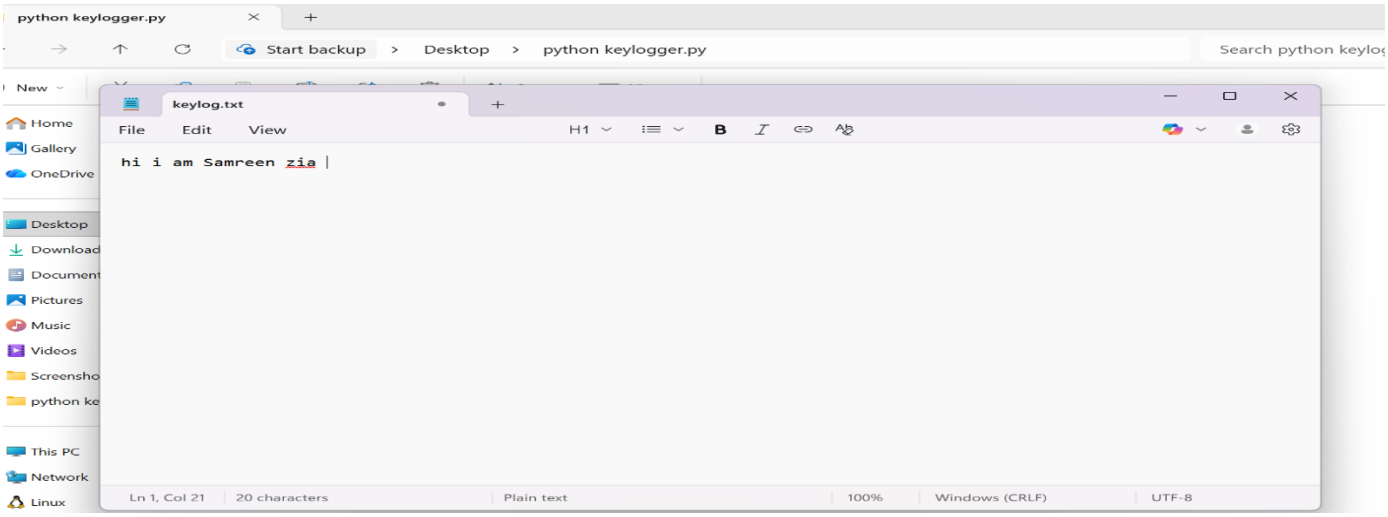
- **Library Installation:** I installed the pynput library via the terminal to enable keyboard event listening.
- **Developing the Script:**
  - I created a listener that triggers the on_press function every time a key is struck.
  - I implemented a try-except block to capture character keys as strings while handling "Special Keys" (like Space or Enter) to prevent script crashes.
  - I set up a stop mechanism in the on_release function to terminate the program gracefully when the **ESC** key is pressed.
- **Logging Mechanism:** The script opens the keylog.txt file in append ("a") mode. This ensures that every new keystroke is added to the end of the file without deleting existing data.



4. Results & Analysis

Based on the live execution of the program:

- **Successful Initialization:** The script was launched via the terminal and began monitoring in the background.
- **Live Testing:** During the simulation, I typed the phrase " hi i am samreen zia" into a notepad/application.
- **Data Verification:** After typing, I pressed the ESC key in the terminal to stop the program. Upon checking the "Desktop Files" folder, I opened keylog.txt and verified that the exact string " hi i am samreen zia" had been captured and stored correctly.
- **Security Risk:** This test confirms that sensitive information (like names, messages, or passwords) is vulnerable if an unauthorized background script is running.



5. Conclusion

The task was successfully completed. By verifying the captured text in the log file, I gained a practical understanding of how input-based attacks work. This reinforces the need for robust antivirus software and cautious installation of third-party applications.